

PAPER_944: Merger Strain Damping

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** `smbh_binary_mergers.py` (MergerStrainDamping) **Calculator:** MergerStrainDampingCalc (CP4 #528) **CVW:** v2.0.0 compliant

Abstract

We quantify gravitational-wave strain damping in SMBH binary mergers due to UQFF buoyancy effects. The total damping factor $D_{\text{total}}(q) = 0.333 + 0.197(1 - q)$ predicts that equal-mass mergers ($q = 1$) retain only 33.3% of the GR strain amplitude, while extreme mass-ratio inspirals approach 53% retention. The UQFF-corrected strain $h_{\text{UQFF}} = h_{\text{GR}} \cdot D_{\text{total}}$ provides a testable prediction for LISA observations.

1. Damping Formula

$$D_{\text{total}}(q) = 0.333 + 0.197 \cdot (1 - q)$$

$$h_{\text{UQFF}} = h_{\text{GR}} \cdot D_{\text{total}}(q)$$

2. Mass-Ratio Dependence

q	D_{total}	Strain Retained	Damping
0.0	0.530	53.0%	47.0%
0.2	0.491	49.1%	50.9%
0.5	0.432	43.2%	56.8%
0.8	0.372	37.2%	62.8%
1.0	0.333	33.3%	66.7%

3. LISA Implications

For a fiducial $h_{\text{GR}} = 10^{-17}$ at $q = 0.5$, the UQFF prediction is $h_{\text{UQFF}} = 4.32 \times 10^{-18}$, well within LISA sensitivity for massive binaries at $z < 1$.

4. Source Data

- **File:** `smbh_binary_mergers.py`
 - **Session:** 213
 - **CP4 Class:** MergerStrainDampingCalc (#528)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. LISA Consortium (2023) – arXiv:2402.07571